1. For the following C statement, what is the corresponding MIPS assembly code? Assume that the variables x , y , and z are saved in $s0, $s1, $s2,$s3. Please include instruction “addi”. (10%)

```
y= ( x+5 ) − (z−10);
```

```
addi  $t0 , $s0 , 5
addi  $t1 , $s2, -10
sub   $s1 , $t0, $t1
```

2. For the following MIPS assembly instructions above, what is a corresponding C statement? (10%)

```
sub  a, b, c;
sll  d, a, 5;
```

```
a=b-c
d=32×a
```

3. Assume that the variables i and j are assigned to register $s2, $s3. Assume that the base address of arrays P and Q are in register $s6 and $s7, respectively. Variable ACC are assigned to register $s5. For the MIPS assembly instruction below, what is the corresponding C statement? Please answer (a) to (e) step by step. (20%)

sll $t0, $s2, 2	
add $t0, $s6, $t0	
sw $s5, 8($t0)	# (a) What is the meaning of these three lines (Please use the variables in the statement of the question if the registers are defined.)
sll $t2, $s3, 2	
add $t2, $s7, $t2	
lw $t0, 0($t2)	# (b) What is the meaning of these three lines
add $s5, $s5, $t0 lw $t0, -4($t2)	# (c) What is the meaning of this line? # (d) What is the meaning of this line?
sub $s5, $s5, $t0	# (e) What is the meaning of this line?

(a) 將 ACC 儲存至 P[i+2]

(b) 將 Q[j] 的值 load 至暫存器 $t0

(c) ACC = ACC + Q[j]

(d) 將 Q[j-1] 的值 load 並更新暫存器 $t0

(e) ACC = ACC − Q[j-1]

4. Translate the following C code to MIPS. Assume that the variables j is assigned to register $s1. Assume that the base address of arrays A and B are in register $s2 and $s3, respectively.(20%)

$$A[j] = B[2j] - B[2j-3]$$

```
        $s1   $s2   $s3
         j     A     B
sll   $t0, $s1, 3
add   $t0, $t0, $s3     # $t0 = B[2j] memory 位置
addi  $t1, $t0, -12     # $t1 = B[2j-3] memory 位置
lw    $t2, 0($t0)
lw    $t3, 0($t1)
sub   $t4, $t2, $t3     # B[2j] - B[2j-3]
sll   $t5, $s1, 2
add   $t5, $t5, $s2     # $t3 = A[j] memory 位置
sw    $t4, 0($t5)       # B[2j] - B[2j-3] 存至 A[j]
```

5. Assume that registers $s0 and $s1 hold the values 0x7FFFFFFF and 0x90000002, respectively.

(a) What is the value of $t7 for the following assembly code? Please express in hexadecimal. (5%)

```
sub $t7, $s0, $s1
```

```
  0111 1111 ..... 1111
+ 0110 1111 ..... 1110
-------------------------
  1110 1111 ..... 1101
```

$t7 : 0xEFFFFFFD #

(b) What is the result in $s2 and $s3 for the following instruction? (10%)

slti $s2, $s1, -1
sltu $s3, $s0, $s1

slti $S2, $S1, -1 signed number 比較

$S1 = 1001 0000 …… 0010 (負數且小於-1)

∴ $S2 = 1

---

sltu $S3, $S0, $S1 unsigned number 比較

∵ 0x7FFFFFFF < 0x90000002 ∴ $S3 = 1

6. Consider the following MIPS loop:

```
LOOP: slt  $t2, $0,   $t1
      beq  $t2, $0,  DONE
      subi $t1, $t1, 1
      addi $s2, $s2, 5
      j      LOOP
DONE:
```

(a) Assume that the register $t1 is initialized to the value 15. What is the value in register $s2 assuming $s2 is initially zero? (5%)

(b) For each of the loops above, write the equivalent C code routine. Assume that the registers $s1, $s2,and $t1are integers A, B, and i, respectively. (10%) (hint:Use while loop or for loop.)

(c) For the loops written in MIPS assembly above, assume that the register $t1 is initialized to the value N. How many MIPS instructions are executed? (10%)

(a) $S2 = 5 × 15 = 75

(b)
```
for (i=15 ; i>0 ; i--) {
    B = B+5;
}
```

(c) 5N + 2

N 減至 0，每一loop 有 5 個指令

當 N ($t1) = 0 時，slt $t2, $0, $t1 不成立，$t2 = 0

beq $t2, $0, Done 成立

跳至 Done，所以多 2 指令